Unit-3

1. What is the purpose of authentication when using the OpenAI API?
2. Name any two vector databases commonly used for storing embeddings.
3. Explain the difference between the completion and chat endpoints in the OpenAI API.
4. Why is JSON output formatting useful when working with API responses?
5. Suppose you are building a chatbot using LangChain. How would you use *chains* and *tools* to integrate external APIs for enhanced functionality?
6. You are retrieving embeddings from HuggingFace Transformers. Show how these embeddings can be stored and searched using FAISS.
7. Compare FAISS and ChromaDB in terms of scalability, search efficiency, and ease of integration with LLM-based applications. Which would you recommend for a large-scale semantic search system, and why?
8. In a multi-agent LangChain setup, discuss the trade-offs between relying solely on API-based hosted models (like HuggingFace) versus using fine-tuned local models.

Unit-4

1. Explain how context retention improves the performance of chatbots.
2. Why is planning important before starting a mini AI project?
3. You are tasked with building a text summarizer using an LLM. Describe the steps you would take to integrate it with Streamlit for a user-friendly interface.
4. Suppose you are developing a personal AI assistant. How would you implement stateful memory to ensure the assistant remembers user preferences across sessions?
5. Compare the challenges of integrating AI applications into web frameworks (like Flask) versus rapid prototyping frameworks (like Streamlit). Which would you choose for a production-grade chatbot, and why?
6. Critically analyze how context retention and UI integration together influence the user experience in an AI-powered translator application.

Unit-5

1. Explain why latency measurement is important in AI application deployment.
2. How does cost estimation relate to token usage in API-based LLMs?
3. You are optimizing a chatbot application. How would you measure latency and identify performance bottlenecks?
4. Suppose your primary model fails to respond in time. How would you implement a fallback strategy using a secondary model?
5. Compare relevance, coherence, and accuracy as evaluation metrics. Which do you think is hardest to measure objectively, and why?
6. In a multi-model orchestration setup, analyse the trade-offs between using a cheaper but less accurate model versus a more costly but highly accurate one.